

Module 1 – Generative AI Essentials

Demo 2 – Explore Vertex AI to Test Text Generation

Problem Statement

Company: Eventora – a company specializing in curated event experiences.

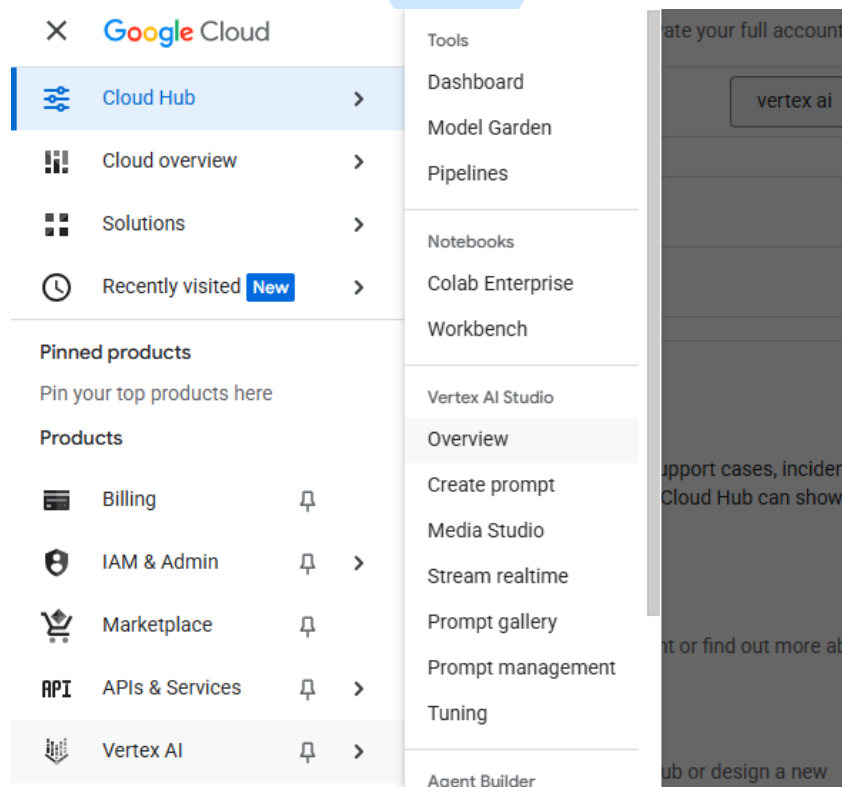
Problem: The marketing team wants to generate appealing event descriptions. However, they need varying tones and output lengths for different channels (web, email, print). The team wants to understand how they can creatively control the AI's behavior using Gemini 2.5 Flash.

Objective

Explore how prompt outputs change with different generation settings like temperature, top-p, and max tokens, using real-world business text generation examples in Vertex AI Studio.

Open Vertex AI Studio

Step 1: From the navigation menu, go to Vertex AI, and click on Overview.



Note: Or search for **Vertex AI API**, and **Enable** API,

← Product details



Vertex AI API

[Google Enterprise API](#)

Train high-quality custom machine learning models with minimal machine learning expertise and...

Enable

Try this API ↗

Step 2: Note if the status of the Vertex AI API is enabled.

← API/Service Details ■ Disable API

To use this API, you may need credentials.

Create credentials



Vertex AI API

Train high-quality custom machine learning models with minimal machine learning expertise and effort.

By Google Enterprise API ⓘ

Service name
aiplatform.googleapis.com

Type
Public API

Status
Enabled

Documentation
[Learn more ↗](#)

Explore
[Try in API Explorer ↗](#)

Step 3: Click on **Try now**, in Vertex AI Studio.

Dashboard

Vertex AI empowers machine learning developers, data scientists, and data engineers to take their projects from ideation to deployment, quickly and cost-effectively. [Learn more about Vertex AI ↗](#)

Enable all recommended APIs



Learn how to use generative AI, Gemini, and more on Vertex AI.

[View tutorials](#)

▼ Show API list



Vertex AI Studio
Prompt, test and tune generative AI models like Gemini.

→ Try now



Model Garden
Browse, customize, and deploy machine learning models. Choose from Google or popular open-source models.

→ Try now

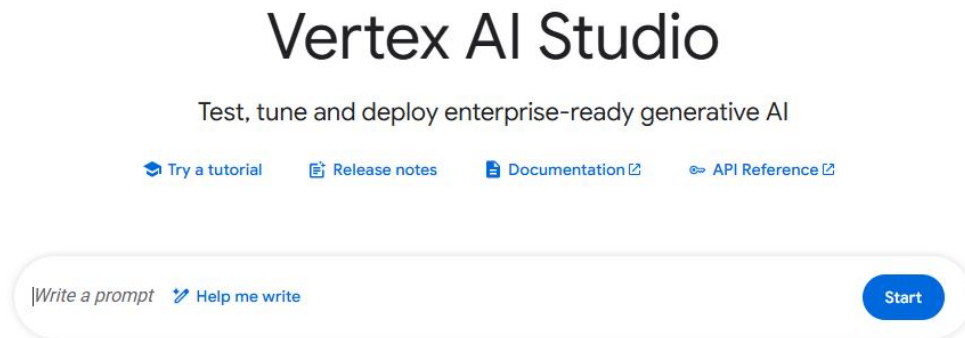


Colab Enterprise
A new notebook experience with enterprise-grade privacy and security. Start coding in a couple clicks.

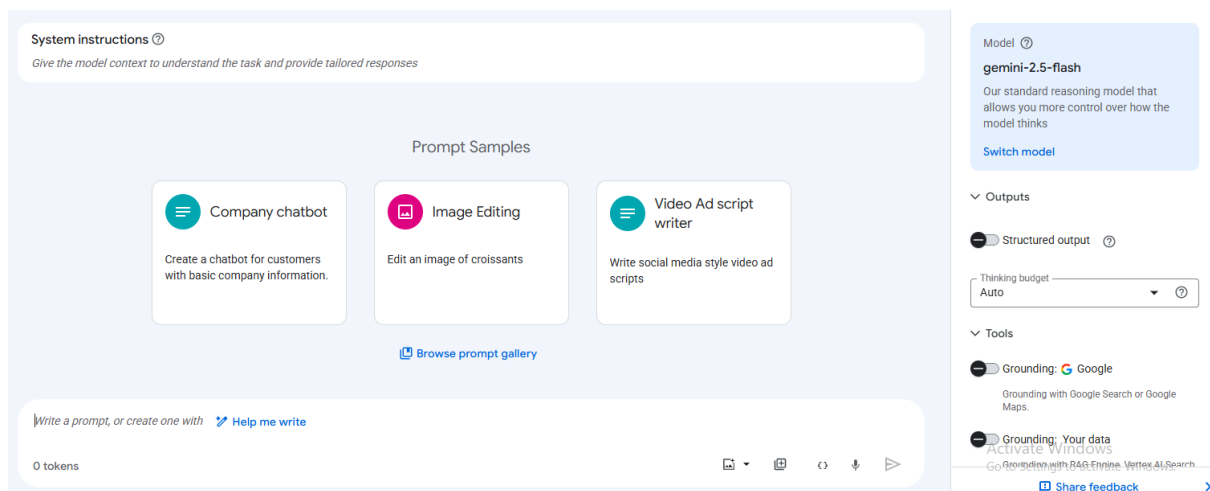
→ Go to Colab Enterprise

Activate Windows
Go to Settings to activate Windows.

Step 4: Click Start.



Step 5: On the top-right corner, choose **Gemini 2.5 Flash** from the model dropdown.

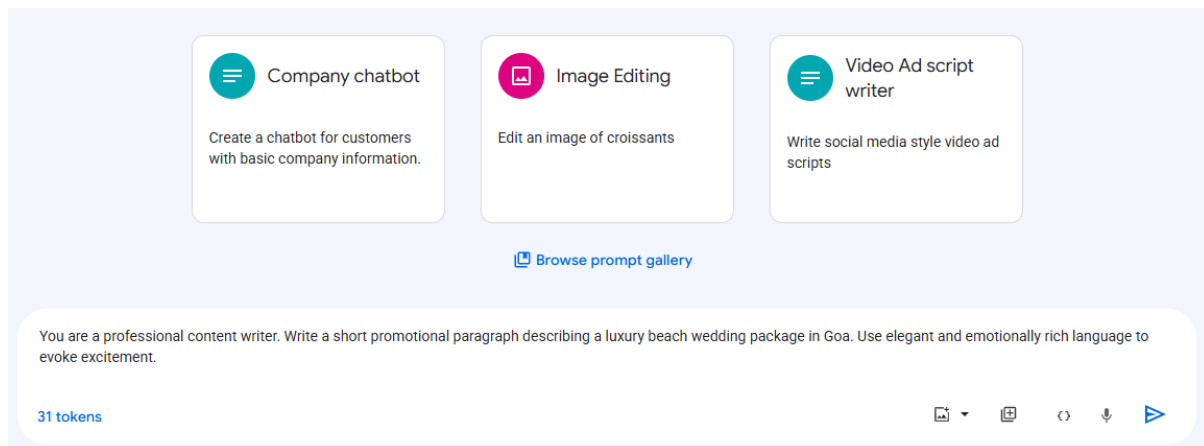


Run Base Prompt (Default Settings)

Step 6: Copy and paste the prompt from below in the chat box.

Prompt: You are a professional content writer. Write a short promotional paragraph describing a luxury beach wedding package in Goa. Use elegant and emotionally rich language to evoke excitement.

Step 7: Keep settings as they are (default temperature, top-p, and tokens)



Step 8: Click on the Send Arrow button.

Step 9: Note the style, tone, and language used.

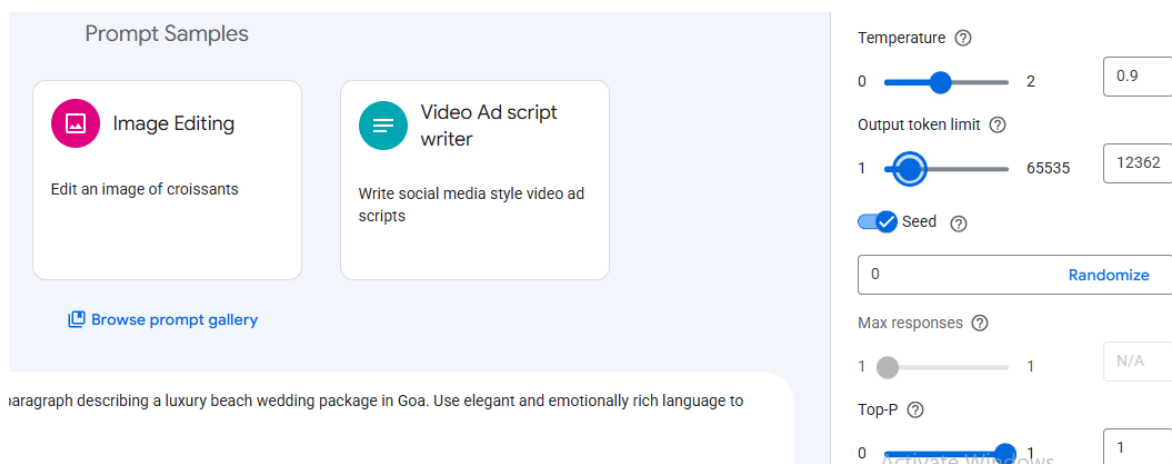
Note: Notice if the language becomes more imaginative, less repetitive, or uses more unique phrasing. You might see playful adjectives or surprising metaphors.



Adjust Temperature to Add More Creativity

Step 10: Set the instructions, under the Advanced setting dropdown menu in the left pane:

- Set **Temperature** = 0.9
- Set **Top-p** = 1.0
- Paste the same prompt again from **step 6**.



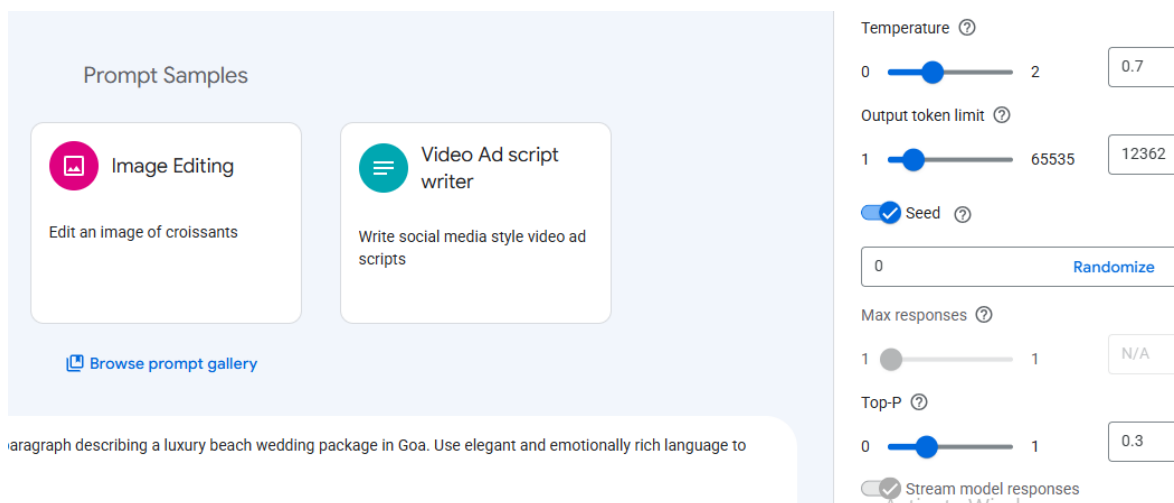
Step 11: Click **Submit**

Step 12: Observe the output. (Note: you can adjust the Max Token limit.)

Adjust Temperature to Add More Creativity

Step 13: Set the instructions, under the Advanced setting dropdown menu in the left pane:

- Set **Temperature** = 0.7
- Set **Top-p** = 0.3
- Paste the same prompt again from **step 6**.



Note: The output should be more predictable and focused, possibly with a simpler or more conservative vocabulary.

Control Output Length – Max Tokens

Step 14: Set the instructions, under the Advanced setting dropdown menu in the left pane

Prompt: You are a travel copywriter. Write a one-paragraph description (max 60 tokens) of a mountain retreat for remote teams, focusing on productivity and relaxation.

- Temperature = 0.6
- Top-p = 1.0
- Max Tokens = 60-100 (Adjust the token limit if the model fails to generate it in the given token limit.)

The screenshot shows the OpenAI Playground interface. On the left, under 'Prompt Samples', there are two cards: 'Image Editing' with the prompt 'Edit an image of croissants' and 'Video Ad script writer' with the prompt 'Write social media style video ad scripts'. Below these is a link 'Browse prompt gallery'. On the right, the settings panel is visible with the following values: Temperature 0.6, Output token limit 12362, Seed checked, Max responses N/A, and Top-P 1.0. A text input field at the bottom contains the prompt: 'ix 60 tokens) of a mountain retreat for remote teams, focusing on productivity and relaxation.'

Step 15: Click **Submit**.

Step 16: Note that the text will now be concise. Useful for chatbot responses, headlines, or meta descriptions.

Combine Settings for Corporate Output

Step 17: Set the instructions under the Advanced setting dropdown menu in the left pane.

- Temperature = **0.3**
- Top-p = **0.2**
- Max Tokens = **40**

Prompt: Describe a professional networking event in under 40 tokens. Keep the tone serious and corporate.

System instructions ⓘ
Give the model context to understand the task and provide tailored responses

USER You are a travel copywriter. Write a one-paragraph description (max 60 tokens) of a mountain retreat for remote teams, focusing on productivity and relaxation.

AI

▼ Thoughts
Expand to show model's summarized thoughts

Boost team productivity and well-being at our mountain retreat. Stunning views, fast Wi-Fi, and serene surroundings offer ideal focus and rejuvenation. Work smarter, relax deeper.

Describe a professional networking event in under 40 tokens. Keep the tone serious and corporate.

1,860 tokens

Settings:

- Temperature ⓘ: 0.3
- Output token limit ⓘ: 60
- Seed ⓘ: 0 (Randomize)
- Max responses ⓘ: 1 (N/A)
- Top-P ⓘ: 0.2
- Stream model responses: ☒ Print responses as they're generated
- Add stop sequence ⓘ

Go to Settings to activate Windows. [Share feedback](#)

Step 18: Click Submit.

Expected Output: A short, precise, no-fluff description. Ideal for enterprise use.

How Each Setting Works (Reference Table)

Setting	Controls	Higher Value Means	Lower Value Means
Temperature	Creativity / randomness	More variation and surprises	More repetitive / safe
Top-p	Probability distribution (focus)	Diverse vocabulary	Narrow, safer word choice
Max Tokens	Output length (in tokens)	Longer paragraphs	Short, precise responses

Conclusion

This demo equips learners with practical skills to shape text outputs using Gemini 2.5 Flash in Vertex AI Studio. With no coding required, they gain full creative and structural control, enabling precise, purposeful content generation for diverse business needs.